Assignment #5

Feature Detection and Matching for Object Detection

Submit By: 11:59pm October 20, 2024
revised October 5, 2024

1 Overview

For this assignment you will be exploring the feature detection and matching algorithms covered in this course. You will choose an object, such as a vase, animal, etc., take several pictures of that object and try to identify that object in each of the pictures using feature detection and matching.

2 Instructions

For this assignment

1. Choose an object to try and detect. Next take three different images of the object if different positions, environments, *etc.* One of these will be your "master" image.
2. Use three different methods of identifying features in an image to find the features for the master image as well as the two other images containing your object.
3. For each of the three feature detection methods used, apply two different feature matching methods and draw the matches. In total, this should give you six different feature/object detection methods to test and compare.
4. For each object detection method, match your master picture to the two other pictures, drawing the matches. Save the output as a screenshot.
5. Finally, submit a write-up, all of your source code, and your twelve outputs containing the drawn matches in a zipped folder.

3 Requirements

Make sure your zipped submission contains:

1. Write-up: Describe what did/didn't work well. Compare the different methods you used.
2. Source code with your name at the top.
3. Your output images with feature matches drawn. Should be 12 images.

4 Submission Instructions

Submit via Canvas or to my email (grwells@uidaho.edu). Most assignments require a write-up and source code, both submitted in a compressed/zipped folder containing all files required to run. Write-ups must be submitted in PDF format. When submitting an assignment via email, please put the course number and the assignment name in the subject line. Submission requirements may be amended in assignment instructions.